# Sparse Matrix Support and Graph Neural Networks in Needle

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Implement sparse-by-dense matrix multiplication on the Needle CPU backend using compressed sparse row (CSR) storage. Train a two-layer GCN on Cora and compare it with the same model using dense Needle operations.

## Division of work

**Member A** Implement CSR storage, CPU kernels, and gradients for stored values and dense inputs. Measure adjacency storage and peak process memory.

**Member B** Build the dense reference tests, Cora loader, GCN, and masked training loss. Run both models and measure time per epoch and test accuracy.

## How we will compare the two implementations

Check both gradients against dense Needle and finite differences. On the fixed Cora graph, start from identical weights with dropout off and compare logits, loss, gradients, and weights after one update. Benchmark on the same CPU with the same split and training budget.

## Milestones

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| **Nov 9 / 10** | Mentor check-in on Nov 9. Discuss scope and agree on the experiments on Nov 10. |
| **Nov 11 / 13** | Build the CPU backend and load Cora by Nov 11. By Nov 13, match sparse and dense outputs and run one dense GCN epoch. |
| **Nov 16** | Submit the course check-in form. Demonstrate the input gradient and test the training mask. |
| **Nov 18 / 20** | Verify both gradients by Nov 18. By Nov 20, compare one GCN update and use pilot timings to set the shared epoch count. |
| **Nov 23** | Finish the Cora runs. Record adjacency bytes, peak process memory, median time per epoch, and accuracy. |
| **Nov 25 / 28** | Record the 90-second video by Nov 25. Submit it by Nov 28. |
| **Dec 3 / 5 / 7** | Draft the Colab report and private code archive by Dec 3, rerun in a fresh session by Dec 5, and submit the report by Dec 7. |

## Questions for the mentor

1. Is CPU sparse multiplication with first-order gradients and a two-layer GCN sufficient for two people?
2. Can we test gradients for stored values separately and keep Cora’s adjacency fixed during training?
3. Do the dense reference and Cora comparison suffice? If time allows, should we vary graph density or add training runs?